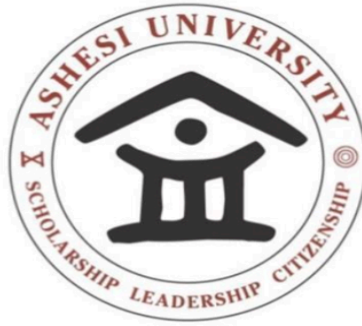




Ashesi University

CS222: Data Structures and Algorithms

Title:

Week 2 Lab Activity (Problem 3 & 4)

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Course: Data Structures and Algorithms

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Problem 3: Checking the math (The sorting performance)

The Casual Answer: Yes, it matches perfectly! Whether you give this Selection Sort algorithm an array that is already perfectly sorted, completely backward, or totally scrambled, it doesn't care. It has no "early exit" button. The outer loop always tracks through every single element, and the inner loop always scans the rest of the array to find the smallest item .

Because it blindly checks everything every single time, it runs the exact same number of basic comparison operations every single time . That's why the code output hits the formula $N(N-1)/2$.

Problem 4: What is java doing in the back?

- After compilation, java returns T into an object. And **no**, the JVM actually doesn't know what's inside at runtime. As far as the running program is concerned, it's just holding an **object**
- Since we put a restriction on it (extends Comparable), Java replaces T with **Comparable** instead of Object. And if it was **<T extends Number & Comparable<T>>**, It would erase to **Number**
- Java complains because Java arrays are strict; they need to know *exactly* what type they are holding at runtime. **The compiler guarantee it's 100% safe** Because we are creating a raw new Node[] array and forcing it to look like a generic Node<T>[] array